

Artificial Intelligence

Lesson 13: Reinforcement Learning I

Value Iteration

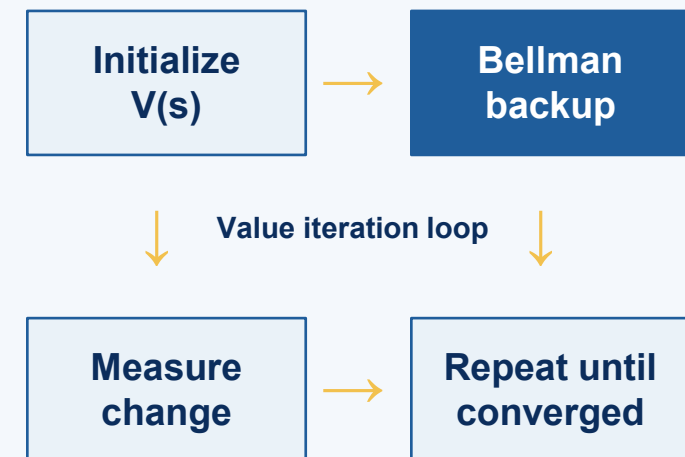
Value iteration improves what we believe about the present using what we currently believe about the future.



Fall 2026

3.1 →	6.5 →	9.0 →	+10
-0.2 ↑	WALL	4.9 ↑	-10
-3.4 →	-0.6 →	1.7 ↑	-1.2 ←

Values become a heat map.



Quiz 3

Individual work, 10 minutes to demonstrate the MDP foundations.

Before you begin

Put away devices
Use only allowed materials
Write clearly and show work
when requested

Focus

L11 MDP components
L12 policy, return, and value
Conceptual distinctions

After quiz

Return to the Gridworld
Values will start changing
Bellman updates begin

After quiz: Where do the values in a Gridworld actually come from?

Today's mission

Policy and value give meaning to behavior over time.

- 1 Explain how value iteration works.**
- 2 Apply a Bellman update to a simple state.**
- 3 Interpret convergence of value estimates.**

Performance target: given transition probabilities, rewards, a discount factor, and current state values, calculate updated action/state values and explain what repeated small changes indicate.

Retrieval: what did the numbers mean?

Previous lesson gave the meaning of value before we compute it.

3.1 →	6.5 →	9.0 →	+10
-0.2 ↑	WALL	4.9 ↑	-10
-3.4 →	-0.6 →	1.7 ↑	-1.2 ←

Policy arrows and state values

Quick check
Is 6.5 an immediate reward?
Is it one lucky trajectory?
What does it summarize?

A value summarizes expected long-term return from a state.

The question L12 left unanswered

We know what value means. Where do the numbers come from?

?	?	?	+10
?	WALL	?	-10
?	?	?	?

Value estimates unknown



3.1 →	6.5 →	9.0 →	+10
-0.2 ↑	WALL	4.9 ↑	-10
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Value estimates after repeated updates

Value iteration fills in the map by repeatedly looking one step ahead.

Math checkpoint: probability-weighted future value

L11 read transition probabilities. L13 uses them.

One action can lead to several futures

Outcome	Probability	Value
Stable	0.75	6
Strained	0.25	2

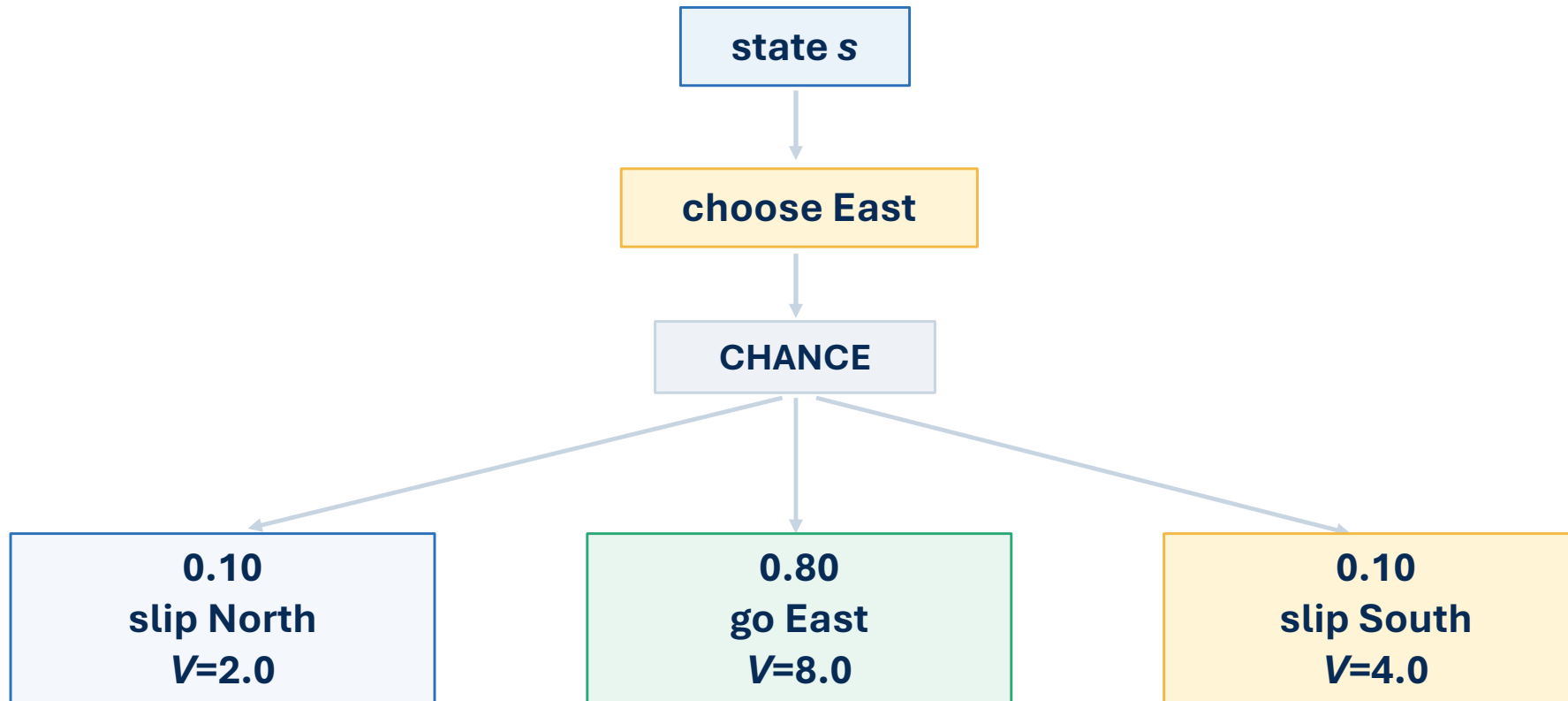
$$0.75(6) + 0.25(2) = 5$$

Multiply each future value by how likely that future is, then add.

This is the probability skill value iteration needs.

A Bellman backup is one-step lookahead

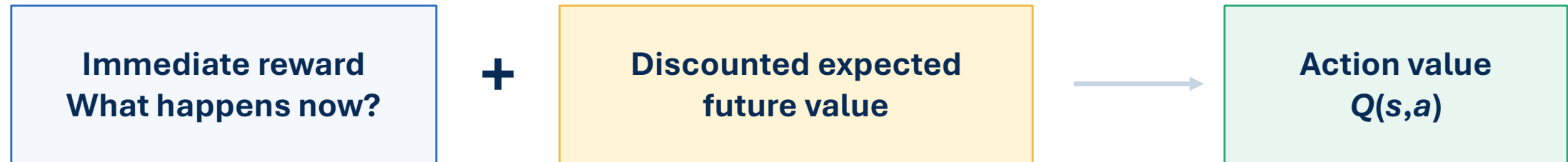
Local structure: agent choice followed by environment uncertainty.



Bellman replaces a deep tree with one-step lookahead plus reusable state values.

The Bellman idea before the equation

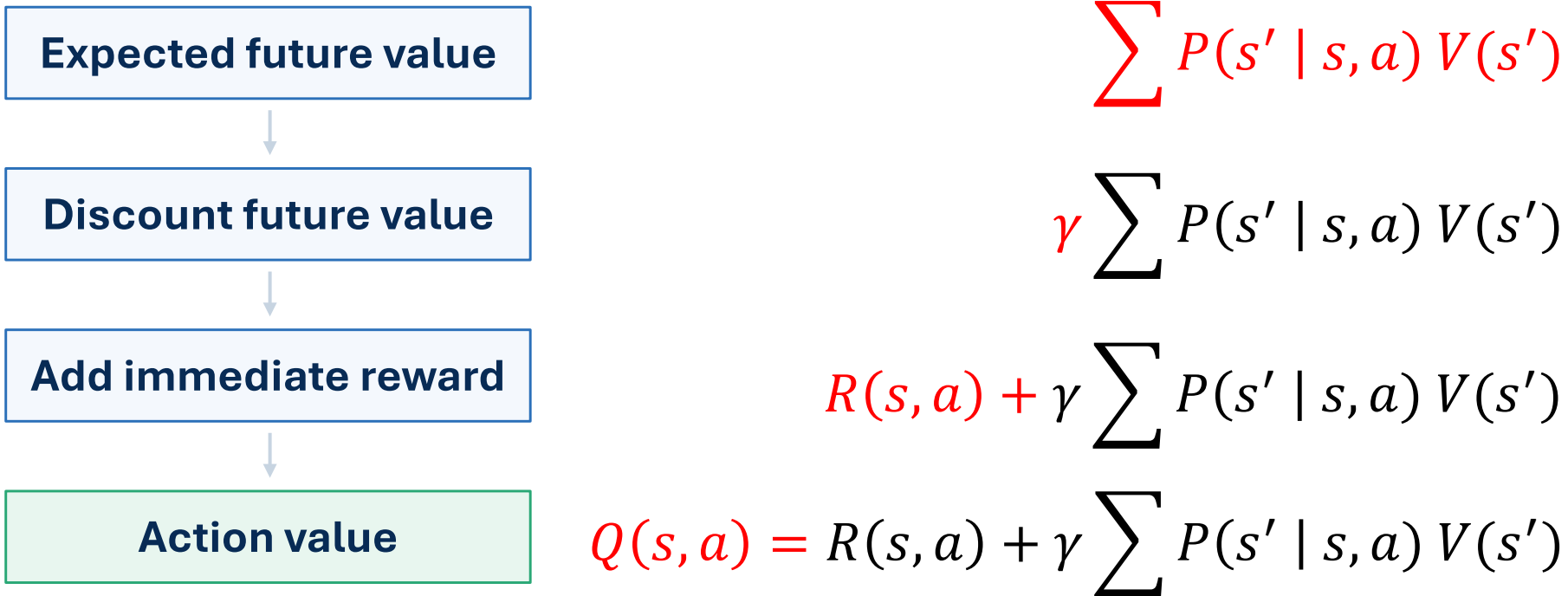
An action has an immediate consequence and leaves us in a future situation.



Question: What do I get now, and what valuable situation am I likely to reach?

Build the Bellman action value

One equation, assembled one idea at a time.



$Q(s, a)$ estimates the long-term value of taking action a in state s .

Read the equation in English

The notation is a compact version of the story.

$$Q(s, a) = R(s, a) + \gamma \sum P(s' | s, a) V(s')$$

Term	Meaning
$Q(s, a)$	value of taking action a here
$R(s, a)$	immediate reward
γ	future-reward weight
$P(s' s, a)$	chance of each next state
$V(s')$	current estimate of next state value

A Bellman backup is immediate reward plus discounted expected future value.

Worked backup: evaluate one action

Zoom into one Gridworld state and one action.

0.0	0.0	0.0	+10
0.0	WALL	0.0	-10
0.0	0.0	0.0	0.0

Current estimates V_0

State	Top row, one cell left of +10
Action	East
Model	0.80 → +10 terminal 0.10 → stay in same cell 0.10 → move south
Reward	living reward = -1 if not terminal

First compute expected future value.
Then add reward and discount.

Step 1: expected future value

Average over environment outcomes.

0.0	0.0	0.0	+10
0.0	WALL	0.0	-10
0.0	0.0	0.0	0.0

$$\sum P(s' \mid s, a) V(s')$$

$$0.80(10) + 0.10(0) + 0.10(0) = 8.0$$

Interpret

If East is attempted, the successor situation has expected future value 8.0 under the current estimates.

Predict

If the south cell becomes more valuable later, this action value may increase.

Do not stop at arithmetic. Explain what the intermediate number represents.

Step 2: discount the future & Step 3: add the reward

Now convert the expected future into an action value.

0.0	0.0	0.0	+10
0.0	WALL	0.0	-10
0.0	0.0	0.0	0.0

$$Q(s, \text{East}) = R + \gamma(\text{expected future value})$$
$$= 0 + 0.90(8.0) = 7.2$$

Meaning

7.2 is the current long-term value estimate for trying East in this state.

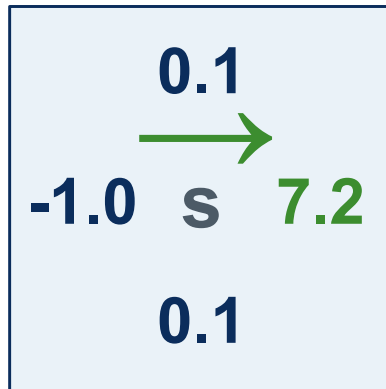
Not meaning

It is not the immediate reward and not a guaranteed outcome.

Action values summarize immediate and future consequences.

A state has choices

A Bellman backup evaluates all available actions, then chooses the best one.



For this state
North: 0.1
East: 7.2
South: 0.1
West: -1.0

0.0	0.0	$\begin{matrix} 0.1 \\ \xrightarrow{\quad} \\ -1.0 \end{matrix} s \begin{matrix} 7.2 \\ 0.1 \end{matrix}$	+10
0.0	WALL	0.0	-10
0.0	0.0	0.0	0.0

$$V_1(s) = \max_a Q(s,a) = 7.2$$

The state value uses the largest action value because the agent chooses the action.

Environment averages; agent maximizes

This is why the Bellman update contains both Σ and $\max$.

Environment uncertainty

The agent does not choose which outcome occurs.

Use transition probabilities.

Average over possible next states.

$$\sum P(s'|s, a)V(s')$$

Agent choice

The agent does choose which action to attempt.

Compare action values.

Select the highest-valued action.

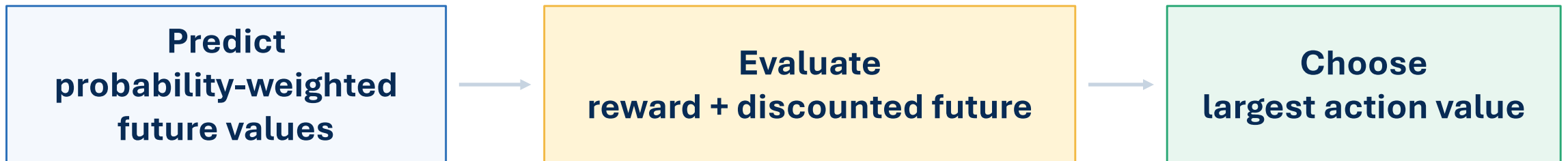
$$\max_a Q(s, a)$$

Sum handles chance. Max handles control.

The Bellman optimality update

Predict → Evaluate → Choose.

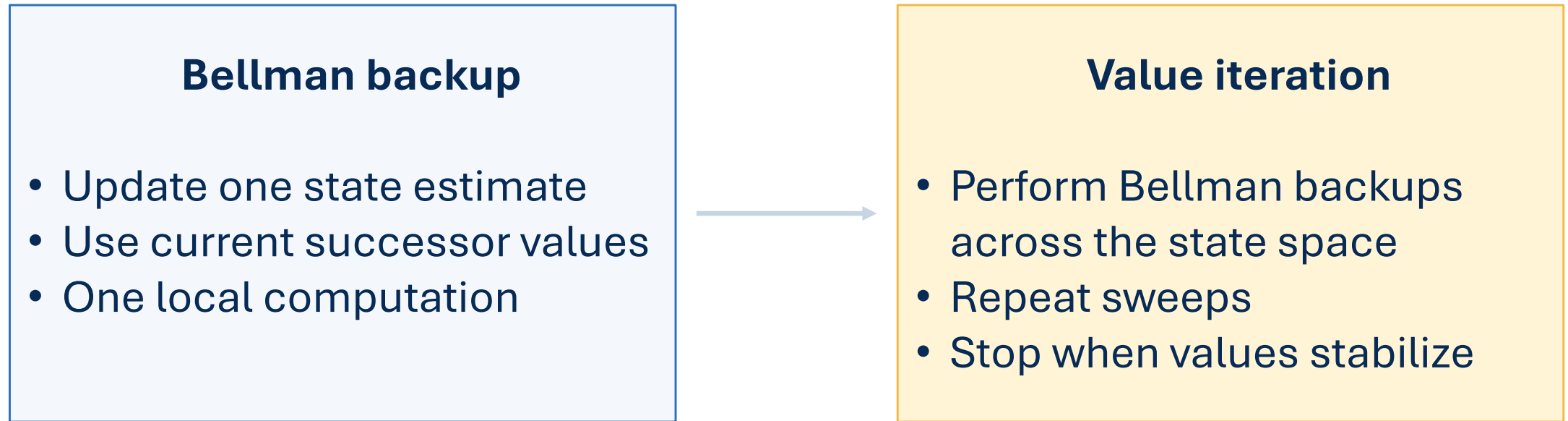
$$V_{k+1}(s) = \max_a [R(s, a) + \gamma \sum P(s' | s, a) V_k(s')]$$



A value iteration sweep applies this update across the state space.

One backup is not value iteration

One local operation becomes an algorithm when repeated across states.



Backup = operation.

Value iteration = repeated backups until stable.

Iteration 0: start with estimates

Initialization is a starting guess, not a claim about truth.

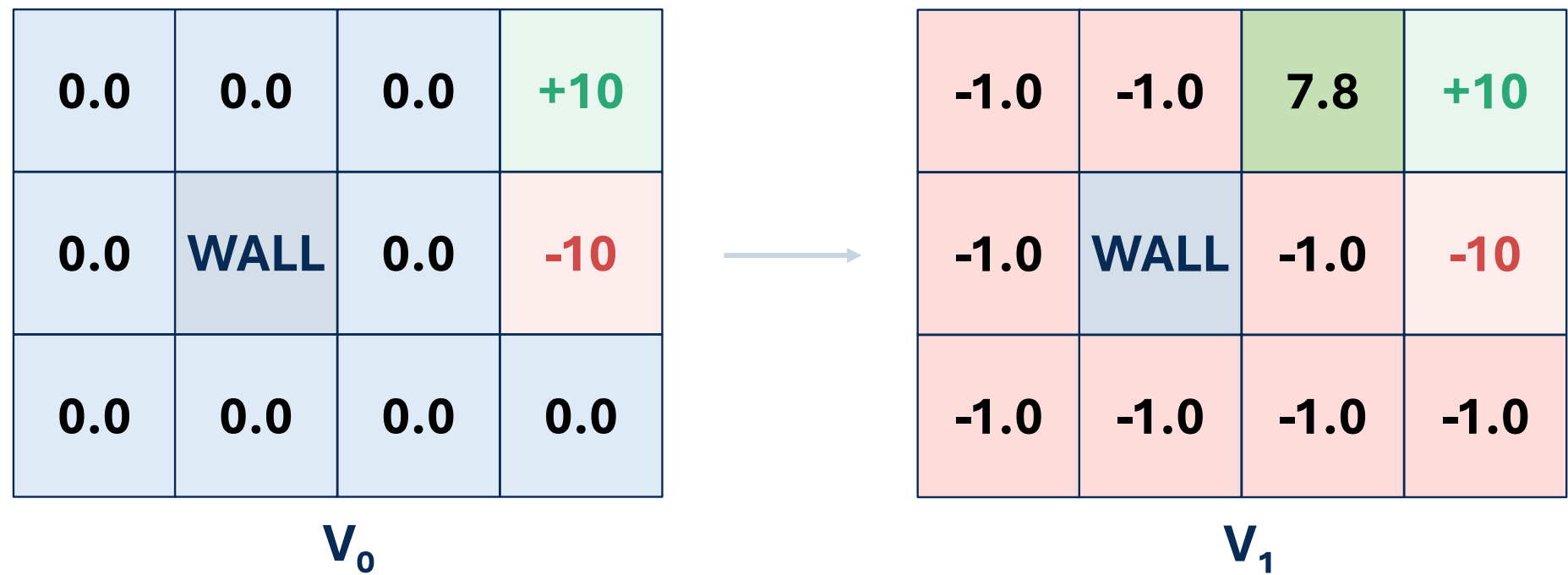
0.0	0.0	0.0	+10
0.0	WALL	0.0	-10
0.0	0.0	0.0	0.0

V_0 : non-terminal state estimates start at 0

Zero means “no future value has propagated yet.”

Iteration 1: reward reaches nearby states

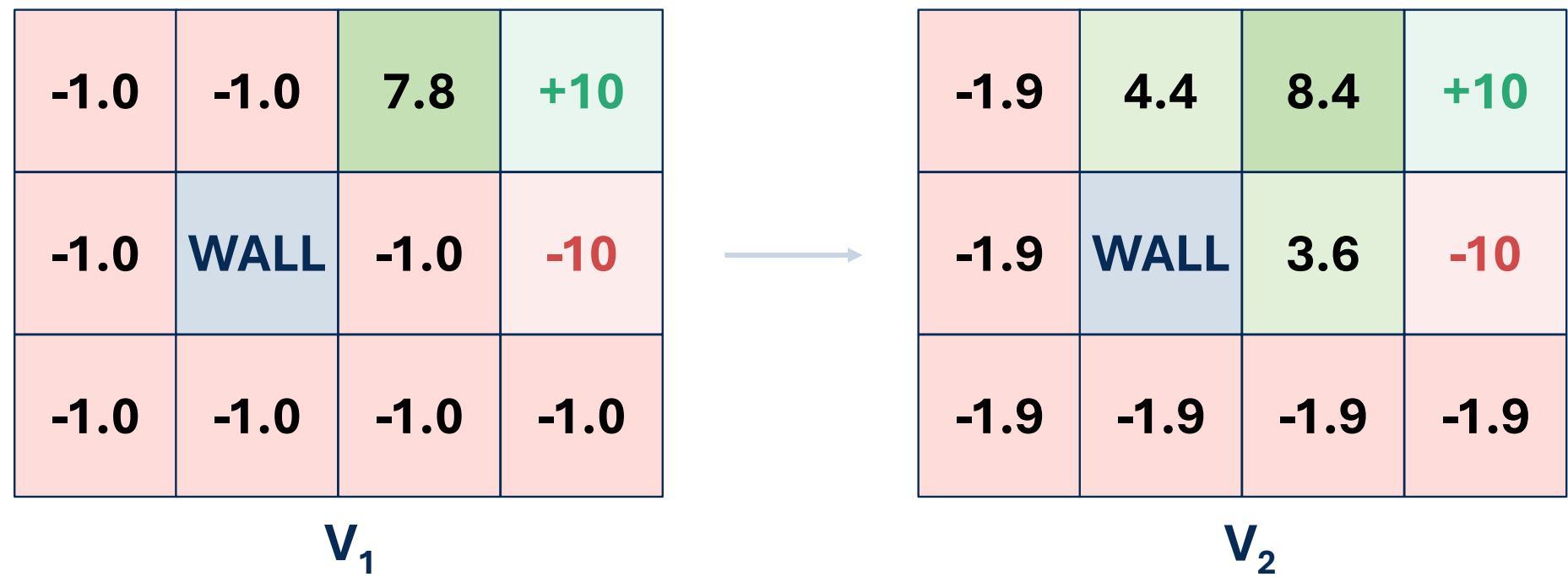
Cells next to terminal outcomes change first.



Why did the cell left of +10 jump first?

Iteration 2: information moves farther

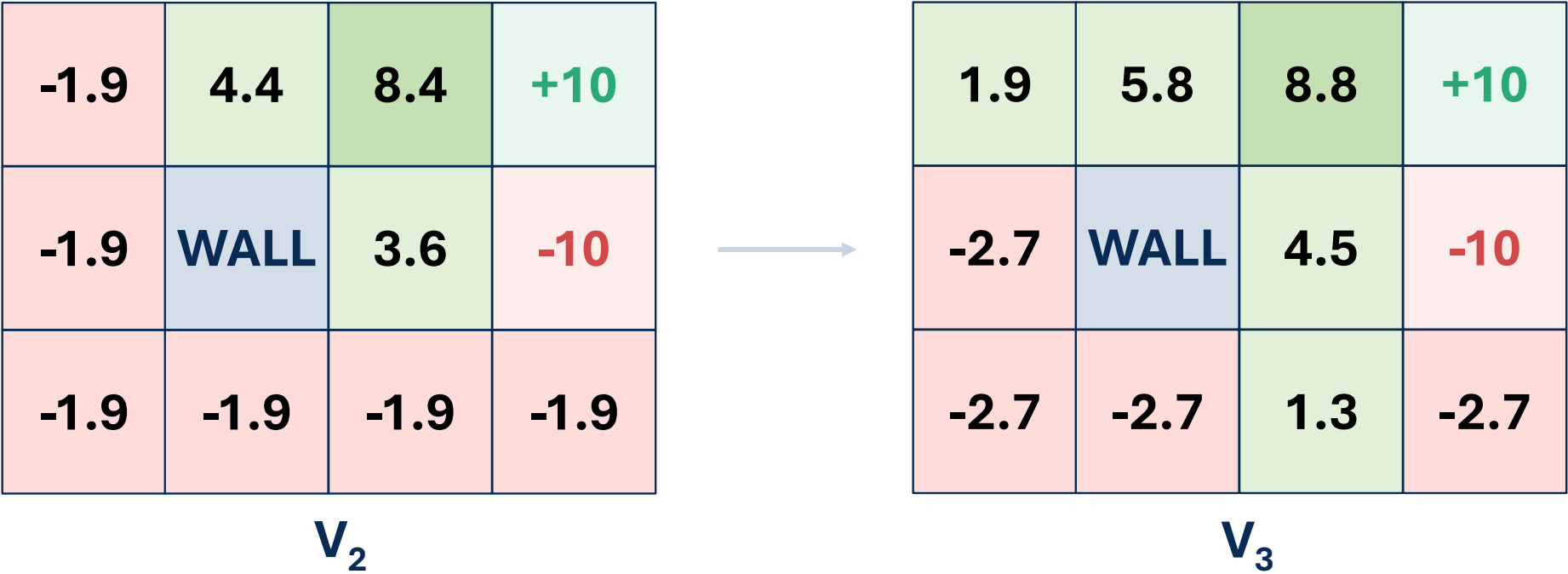
States that can lead to valuable states become valuable too.



The value of the future is propagating backward through the grid.

Iteration 3: long-term consequences spread

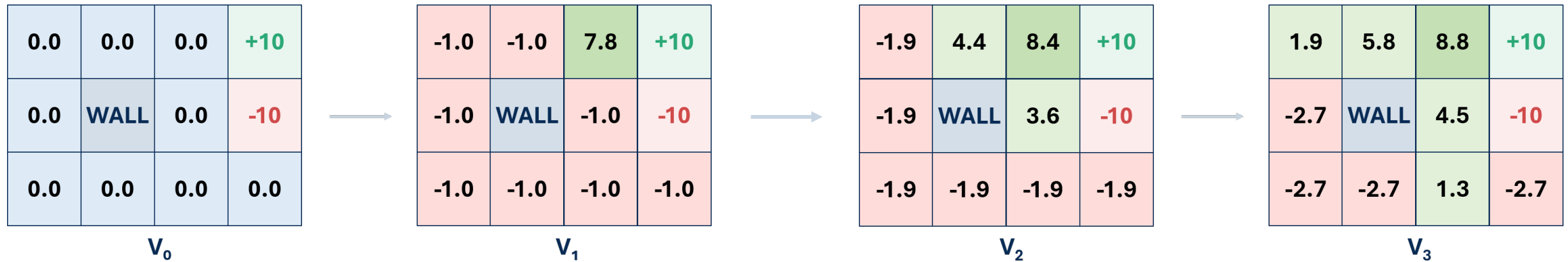
Before revealing a value, predict which cells should improve next.



Predict: if a successor state becomes more valuable, what happens to states that can reach it?

Value propagates backward through the state space

The heat map tells the story of repeated Bellman backups.



Each sweep lets future reward influence states farther away.

Values are not rewards

A state can have low immediate reward and high long-term value.

3.1	6.5	9.0	+10
-0.2	WALL	4.9	-10
-3.4	-0.6	1.7	-1.2

Example state

Immediate reward

Moving through ordinary cells still costs about -1 .

State value

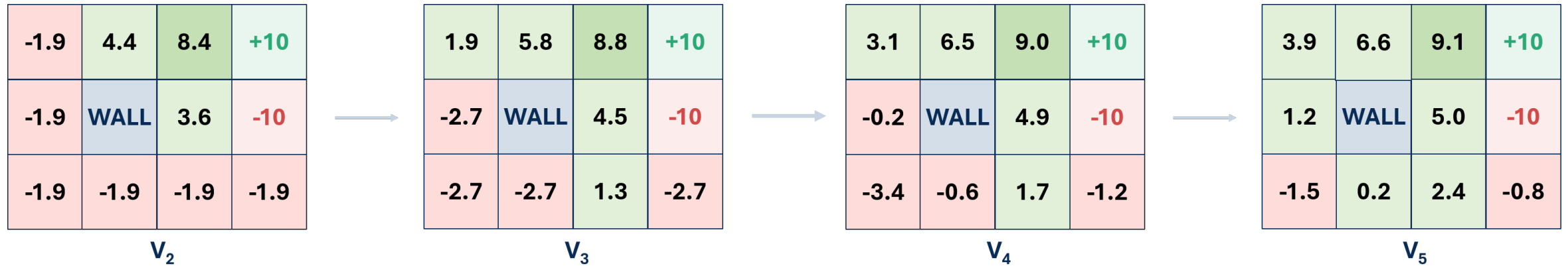
This state can lead toward $+10$, so its long-term value is positive.

Reward answers “what happened now?”

Value answers “what should I expect from here?”

What convergence looks like

The estimates change a lot early and only a little later.



A sample cell: 4.4 → 5.8 → 6.5 → 6.6

Successive heat maps become less different.

Measure change with Δ

Convergence needs a stopping signal.

$$\Delta = \max_s |V_{k+1}(s) - V_k(s)|$$

Interpretation

Δ is the largest state-value change in the latest sweep.

Example

If $\Delta = 0.0007$ and $\theta = 0.001$, stop because $\Delta < \theta$.

Small Δ means the repeated updates are no longer changing the values much.

Convergence means stability, not magic

Values usually do not become exactly unchanged in finite time.

Not convergence

- Every value must be exactly identical
- The agent has experienced the environment
- Rewards disappear
- The math becomes deterministic

Convergence

- Value estimates are sufficiently stable
- Further sweeps change little
- The tolerance tells us when to stop
- The resulting values support action choice

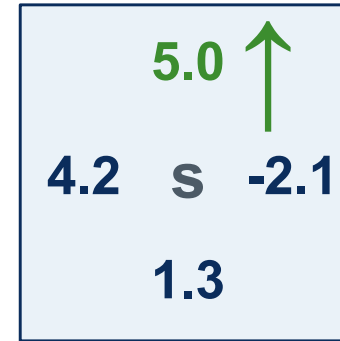
We stop when the values are stable enough for the chosen tolerance.

Values reveal a greedy policy

Once values stabilize, choose the action with the highest Bellman action value.

3.9 →	6.6 →	9.1 →	+10
1.2 ↑	WALL	5.0 ↑	-10
-1.5 →	0.2 →	2.4 ↑	-0.8 ←

Stabilized values + greedy arrows



$$\pi(s) = \arg \max_a Q(s, a)$$

The arrow is not guessed. It comes from comparing action values.

Value iteration is model-based

Every backup used a model of how actions may turn out.

States
S

Actions
A

Transitions
 $P(s'|s, a)$

Rewards
R

**If transition probabilities are unknown,
this exact Bellman expectation calculation cannot be performed.**

That question will return in Q-learning.

Practice: Bellman update sprint

Show substitution, compute, compare, interpret.

**action: Reserve
at state: Z**

immediate reward: +2

70% → X, $V(X)=5$

30% → Y, $V(Y)=1$

$\gamma = 0.6$

$$Q(Z, \text{Reserve}) = 2 + 0.6[0.7(5)+0.3(1)]$$
$$= 4.28$$

Compare

If $Q(Z, \text{Alternate})=4.60$, which action is greedy?
What does 4.28 mean?

Compute → Interpret → Predict.

Practice: diagnose Bellman mistakes

Which statements are correct? Which misconception do they reveal?

A

Choose the most likely next state and ignore the others.

Wrong: use expected value.

B

Choose the action with the largest immediate reward.

Wrong: compare Bellman action values.

C

During a synchronous sweep, use V_k values to compute V_{k+1} .

Correct for our convention.

D

Tiny value changes suggest convergence if below the threshold.

Reasonable.

Common errors usually come from confusing chance, control, and time.

Practice: update from a transition table

Use the transition table to compute action values and update the state value.

Reminder

$$Q_k(s,a) = R(s,a) + \gamma \sum P(s'|s,a)V_k(s')$$

Then update with $V_{k+1}(s) = \max_a Q_k(s,a)$

Current state: A

$\gamma = 0.7$

Action a	Reward	Next state s'	$P(s' A,a)$	Current value $V_k(s')$
East	+1	X	0.8	5
East	+1	Y	0.2	1
South	0	Z	0.6	4
South	0	Y	0.4	1

Tasks

1. Compute $Q_k(A, \text{East})$.

2. Compute $Q_k(A, \text{South})$.

3. Update $V_{k+1}(A) = \max_a Q_k(A,a)$.

4. Which action is greedy at state A?

5. What does $V_{k+1}(A)$ mean in words?

Assume the transition table above contains all nonzero outcomes for state A.

Compute → compare → update.

Read value iteration

Map the code to the visual process.

```
initialize V
repeat:
  delta = 0
  for each state s:
    compute return for every action
    V_new[s] = largest action value
    update delta
  V = V_new
until delta is small
```

Code idea	Visual meaning
compute return	Predict + evaluate
largest action value	Agent takes MAX
update delta	Measure change
repeat	Propagate farther

Question to ask: which part handles chance, and which part handles choice?

Exit ticket: rebuild value iteration

Use the concept chain, not just the formula.

Expected future value combines _____.

Bellman action value adds _____.

$V(s)$ uses the _____ action value.

Repeated sweeps allow value to _____.

Very small changes suggest _____.

Preview L14: What if we explicitly hold a policy fixed, evaluate it, then improve it?